Nonuniqueness of a quadratic BSDE with all exponential moments

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Abstract

We construct a scalar quadratic BSDE with a predictable generator, independent of the value variable and satisfying a uniform quadratic Lipschitz bound, that has a continuum of distinct solutions with the same terminal value. Each constructed solution has a square-integrable integrand and an integrable exponential of its running absolute maximum at every positive order. The generator is neither convex nor concave. Thus the nonconvex uniqueness question has a negative answer even after imposing Lipschitz continuity in the value variable.

Problem 5.5 of Fan, Hu and Tang [1] asks about uniqueness for nonconvex quadratic BSDEs with all exponential moments. Its literal formulation does not control the dependence on y. The construction below is independent of y, so it also answers the question that imposes a uniform Lipschitz bound in y and a bounded generator at the origin. Class (D) means uniform integrability over all stopping times bounded by T.

Theorem 1. *For every $T > 0$, on the usual augmented natural filtration of a one-dimensional Brownian motion B, there exist a predictable generator $g(t, \omega, z)$ and a terminal value ξ such that, for all $z, z' \in \mathbb{R}$,*

$$g(t, 0) = 0, \qquad |g(t, z)| \le z^2, \qquad |g(t, z) - g(t, z')| \le 4(1 + |z| + |z'|)|z - z'|. \tag{1}$$

Every function $g(t, \omega, \cdot)$ is continuous, neither convex nor concave. The equation

$$Y_t = \xi + \int_t^T g(s, Z_s)\, ds - \int_t^T Z_s\, dB_s \tag{2}$$

has a solution (Y^c, Z^c) for every $c \in [0, 1]$, with distinct initial values, and

$$\mathbb{E} \int_0^T |Z_s^c|^2 ds < \infty, \qquad \mathbb{E} \exp\left(\mu \sup_{0 \le t \le T} |Y_t^c| \right) < \infty \quad \text{for every } \mu > 0. \tag{3}$$

In particular, $\mathbb{E} e^{\mu |\xi|} < \infty$ and $e^{\mu |Y^c|}$ is of class (D) for every $\mu > 0$.

A sequence of stopped excursions. Put $s_k = T(1 - 2^{-k})$ for $k \ge 0$. Leg $k \ge 1$ occupies $[s_{k-1}, s_k)$, has sign $\sigma_k = (-1)^{k+1}$, and belongs to cycle $n = \lceil k/2 \rceil$. Set

$$a_n = \log(n + 1), \qquad h_k(t) = \frac{1}{s_k - t}, \qquad L_t^k = \sigma_k \int_{s_{k-1}}^t h_k(s)\, dB_s - \int_{s_{k-1}}^t h_k(s)^2 ds.$$

Extend h_k by zero outside its leg. Let τ_k be the first exit of L^k from $(-a_n, a_n)$. The deterministic clock $\int_{s_{k-1}}^t h_k^2$ tends to infinity as $t \uparrow s_k$. In clock time L^k is a Brownian motion with drift -1, so $\tau_k < s_k$ almost surely. Call the leg successful if $L_{\tau_k}^k = a_n$.

Only legs whose predecessors were all successful are used. Write A_k for this event and define the predictable process

$$q_t = \sum_{k\geq 1} \mathbf{1}_{A_k} \mathbf{1}_{\{s_{k-1}<t\leq \tau_k\}} h_k(t), \qquad X_t = \int_0^t q_s \, dB_s - \int_0^t \sigma(s) q_s^2 ds, \tag{4}$$

where $\sigma(s) = \sigma_k$ on leg k and endpoint values are immaterial. The event A_k is known at time s_{k-1}, and the stopped interval indicators are predictable. Finiteness of the energy in (4) will be verified below; equivalently, one may first construct these processes up to any finite number of legs.

The first leg in a cycle starts at $X = 0$ and, on success, ends at $X = a_n$. The second starts at a_n and, on success, returns to zero. A failed first leg ends at $-a_n$; a failed second leg ends at $2a_n$. After a failure, $q = 0$ forever. Between a successful exit and the next leg, X is constant.

For Brownian motion with drift -1 started at zero in $(-a, a)$, the probability of exiting at a is

$$p(a) = \frac{1}{1 + e^{2a}}.$$

Indeed, apply Itô's formula and bounded optional sampling to the harmonic function $\frac{e^{2(x+a)}-1}{e^{4a}-1}$. The exit time S in clock time also satisfies $\mathbb{E}S \leq a$: at $S \wedge m$, the stopped process lies in $[-a, a]$ and has expectation $-\mathbb{E}(S \wedge m)$, and then monotone convergence applies.

Thus each used leg in cycle n has conditional success probability $p_n = (1 + (n+1)^2)^{-1}$. Independence of Brownian increments on the successive deterministic intervals gives, for the cycle N of the first failure,

$$\mathbb{P}(N \geq n) = \prod_{j<n} p_j^2 \leq \frac{1}{(n!)^4}, \qquad \mathbb{E}\int_0^T q_t^2 dt \leq 2\sum_{n\geq 1} \frac{a_n}{(n!)^4} < \infty. \tag{5}$$

In particular $N < \infty$ almost surely, and

$$\sup_{t\leq T} |X_t| \leq 2a_N, \qquad \mathbb{E}e^{\mu \sup |X|} \leq \sum_{n\geq 1} \frac{(n+1)^{2\mu}}{(n!)^4} < \infty. \tag{6}$$

All paths under consideration stop strictly before T. The null event of infinitely many successes can be discarded using the completed filtration.

A bounded perturbation with zero terminal value. Define

$$F_n(x) = \frac{1 - e^{-2(x+a_n)}}{1 - e^{-4a_n}}, \qquad r_n = F_n(0) = \frac{1}{1 + e^{-2a_n}}, \qquad d_k = \prod_{j=k}^{\infty} r_{\lceil j/2 \rceil}, \qquad \eta = \frac{1}{16}.$$

These products are positive: $\sum_n \log(1 + (n+1)^{-2}) < \infty$. They satisfy $d_k = r_n d_{k+1}$ when $n = \lceil k/2 \rceil$. On $[-a_n, a_n]$,

$$F_n(-a_n) = 0, \quad F_n(a_n) = 1, \quad F_n'' = -2F_n', \quad 0 < F_n' \leq \frac{32}{15}. \tag{7}$$

During a used leg k of cycle n, up to its exit, set

$$D_t = \eta d_{k+1} F_n(L_t^k), \qquad K_t = \eta d_{k+1} \sigma_k h_k(t) F_n'(L_t^k).$$

Hold D constant between legs, set it to zero forever after failure, and put $K = 0$ outside the used portions of legs. At the beginning of leg k, $D = \eta d_k$; on success it becomes ηd_{k+1}. Thus the pieces join continuously. We have

$$D_0 = \eta d_1 > 0, \qquad D_T = 0, \qquad 0 \leq D \leq \eta, \qquad |K_t| \leq \frac{2}{15} q_t. \tag{8}$$

Itô's formula, using the drift -1 of L^k and (7), gives the global identity

$$dD_t = K_t\,dB_t - 2\sigma(t)q_tK_t\,dt. \tag{9}$$

All terms are integrable by (5) and (8). This construction uses no change of probability measure.

The generator and the solutions. Let

$$G(u) = 2\big(1 - |u - \tfrac{3}{2}|\big)_+.$$

This tent function is supported on $[1/2, 5/2]$, is 2-Lipschitz, satisfies $0 \le G(u) \le u^2$, and equals $2u - 1$ on $[1/2, 3/2]$. Define

$$g(t,z) = \begin{cases} \sigma(t)q_t^2 G(z/q_t) + G(-z), & q_t > 0, \\ G(-z), & q_t = 0. \end{cases} \tag{10}$$

The coefficient q was fixed before choosing a solution parameter. Hence this is one predictable generator for all the solutions below. It does not depend on y.

The two terms in (10) have disjoint positive and negative supports, so $|g(t,z)| \le z^2$ and $g(t,0) = 0$. At points of differentiability, the scaled tent has derivative of magnitude at most $2q_t$, supported where $z \ge q_t/2$; its derivative is therefore bounded by $4|z|$. The other tent has derivative bounded by 2. Integration along the interval from z to z' proves (1). Each $g(t,\omega,\cdot)$ is compactly supported and nonzero on the negative axis. A nonzero compactly supported function on $\mathbb{R}$ can be neither convex nor concave.

For $c \in [0,1]$ put

$$Y^c = X + cD, \qquad Z^c = q + cK, \qquad \xi = X_T.$$

Where $q > 0$, equations (8) give

$$\frac{Z^c}{q} \in [13/15, 17/15] \subset [1/2, 3/2], \qquad g(t, Z^c) = \sigma q^2 + 2c\sigma qK.$$

Where $q = 0$, both Z^c and $g(t, Z^c)$ vanish. Combining (4) and (9) now yields

$$dY_t^c = Z_t^c\,dB_t - g(t, Z_t^c)\,dt.$$

Since $D_T = 0$, integration gives (2) with the same ξ for every c. The identities hold simultaneously for all times by continuity, including T, since only finitely many legs are used almost surely.

Finally, $|Z^c| \le (17/15)q$ and $\sup|Y^c| \le 2a_N + \eta$. Equations (5)–(6) prove (3). Each stopped exponential is dominated by the integrable random variable $e^{\mu \sup|Y^c|}$, which proves class (D). The initial values $Y_0^c = c\eta d_1$ are distinct. This proves Theorem 1. □

Scope. The generator is random and time-dependent, as permitted in the stated problem. Taking $g(t,y,z) = g(t,z)$ satisfies any positive uniform Lipschitz bound in y. The example therefore meets the hypotheses with $\delta = 1$ and $C = 4$. In higher Brownian dimension, use the first coordinate and replace $g(t,z)$ by $g(t,z_1)$.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] S. Fan, Y. Hu and S. Tang. *1D nonlinear backward stochastic differential equations: a unified theory and applications*. arXiv:2309.06233v2, 11 February 2026, §1, equation (3.12), and Problem 5.5. `https://arxiv.org/html/2309.06233v2`.